

Fractions, Decimals and Percentages — Mark schemes

PAPERS A TO D • 30 MARKS EACH • TUTOR COPY

Marks in square brackets show how the total is split. Where a question is worth more than one mark, award method marks even when the final answer is wrong — and take them away for a correct answer with no working only if the question said "show your working". The notes under each answer flag the specific mistake that question was built to catch.

Paper A — Medium

30 MARKS • NON-CALCULATOR

- 1** Row 1: **0.75** and **75%**. Row 2: **7/20** and **35%**. Row 3: **3/5** and **0.6**. [6]
One mark per correct box. 35/100 not simplified to 7/20 loses that one box only.
- 2** (a) $45/100 = \mathbf{9/20}$ [2]
 (b) $24/100 = \mathbf{6/25}$
- 3** (a) $7 \div 8 = \mathbf{0.875}$ [2]
 (b) $5 \div 16 = \mathbf{0.3125}$
- 4** (a) $12/20 + 5/20 = \mathbf{17/20}$ [4]
 (b) $5/6 - 2/6 = 3/6 = \mathbf{1/2}$
 (c) $18/30 = \mathbf{3/5}$
 (d) $3/4 \times 5/2 = 15/8 = \mathbf{1\ 7/8}$
Award the mark for a correct unsimplified answer only if the question did not ask for simplest form — here it did.
- 5** (a) **120** [3]
 (b) **9**
 (c) **30**
(c) is 1/8 of 240. Halving three times is quicker than any other route.
- 6** Converting all four to decimals: 0.7, 0.6, 0.65, 0.667 **[1]** [2]
 Order: **3/5, 65%, 2/3, 0.7 [1]**
The second mark requires the original forms. Answering 0.6, 0.65, 0.667, 0.7 loses it.
- 7** 1200×0.8 **[1]** [2]
= 960 rupees [1]
Finding 240 and stopping scores 1 — that is the discount, not the sale price.
- 8** 40% of 30 = 12 wear glasses **[1]** [2]
 $30 - 12 = \mathbf{18 [1]}$
Or 60% of 30 directly. Answering 12 scores 1: the question asked who does not.
- 9** (a) $500 \div 4 = 125$, so $3 \times 125 = \mathbf{375\ g [2]}$ [3]
 (b) $500 - 375 = \mathbf{125\ g [1]}$
- 10** (a) $18/24 = \mathbf{3/4}$ [2]
 (b) **75%**
- 11** 8000×1.05 **[1]** [2]
= 8400 rupees [1]

Paper B — Medium

30 MARKS • NON-CALCULATOR

- 1** (a) **13/5** [2]
(b) **4 1/4**
- 2** (a) $3/2 + 11/4 = 6/4 + 11/4 = 17/4 = 4 \frac{1}{4}$ [4]
(b) $10/3 - 11/6 = 20/6 - 11/6 = 9/6 = 1 \frac{1}{2}$
(c) $5/4 \times 12/5 = 60/20 = 3$ [2]
In (c), one mark for converting both mixed numbers to improper fractions, one for the answer.
- 3** Decimals: 0.46, 0.444, 0.43, 0.45 [1] [2]
Order: **43%, 4/9, 9/20, 0.46** [1]
- 4** 250×1.12 [1] [2]
= 280 [1]
- 5** 640×0.85 [1] [2]
= 544 [1]
- 6** Change = $800 - 680 = 120$ [1] [3]
 $120 \div 800$ [1]
= 15% decrease [1]
Dividing by 680 instead of 800 gives 17.6% — the single most common error in this topic. Award the first mark only.
- 7** Change = $4600 - 4000 = 600$ [1] [3]
 $600 \div 4000$ [1]
= 15% increase [1]
- 8** $320 \div 8 = 40$, so 3×40 [1] [2]
= 120 [1]
- 9** $5/8$ of 400 = **250** [1] [3]
60% of 420 = **252** [1]
60% of 420 is larger (by 2) [1]
The third mark is for the conclusion in words. Two correct numbers with no statement scores 2.
- 10** 68% of 250 = 170 full [1] [2]
 $250 - 170 = 80$ empty [1]
Or 32% of 250 directly.
- 11** $375/1000 = 3/8$ [1] [2]
37.5% [1]
- 12** Boys = $40 - 24 = 16$ [1] [3]
 $16/40$ [1]
= 40% [1]
Answering 60% means the student found the girls. That scores nothing beyond the method mark.

Paper C — Hard

30 MARKS · NON-CALCULATOR

- 1** (a) **Terminates**, 0.1875 ($16 = 2 \times 2 \times 2 \times 2$) [3]
(b) **Recurs**, 0.41666... (12 has a factor of 3)
(c) **Terminates**, 0.28 ($25 = 5 \times 5$)
Both halves needed for each mark. A fraction terminates only when the denominator, in simplest form, has no prime factors other than 2 and 5.

- 2** The multiplier for a 20% increase is 1.2 **[1]** [3]
 Original = $96 \div 1.2$ **[1]**
 = **80 rupees [1]**
Taking 20% off 96 gives 76.80 and scores zero. You must divide, not subtract.
- 3** The multiplier for a 25% decrease is 0.75 **[1]** [3]
 Original = $180 \div 0.75$ **[1]**
 = **240 rupees [1]**
- 4** (a) Starting from 100: $100 \times 1.1 = 110$, then $110 \times 0.9 = 99$, which is not 100 **[2]** [3]
 (b) $1.1 \times 0.9 = 0.99$, so an overall **1% decrease [1]**
Any starting value is acceptable in (a). The second 10% is taken off a larger number than the first was added to, so the two never cancel.
- 5** $3/5$ of 450 = 270 **[1]** [3]
 $2/3$ of 270 **[1]**
 = **180 [1]**
Also acceptable: $2/3 \times 3/5 = 2/5$, then $2/5$ of 450 = 180. Full marks for either route.
- 6** (a) Shop A: $2400 \times 0.7 = 1680$ **[1]** [4]
 Shop B: $2400 \times 0.75 = 1800$, then $1800 \times 0.9 = 1620$ **[1]**
 (b) **Shop B is cheaper [1]**, by $1680 - 1620 = 60$ rupees **[1]**
Adding 25% and 10% to get 35% off is the trap. Successive percentage changes multiply.
- 7** $5/8$ of 640 = 400 **[1]** [3]
 15% of 400 **[1]**
 = **60 [1]**
- 8** Losing 12% leaves 88%, so the multiplier is 0.88 **[1]** [3]
 $4.4 \div 0.88$ **[1]**
 = **5 kg [1]**
Adding 12% to 4.4 gives 4.93 and scores at most the first mark.
- 9** Recognising a single recurring digit gives ninths **[1]** [2]
 = **4/9 [1]**
Worth memorising: $0.111... = 1/9$, so $0.444... = 4/9$ and $0.777... = 7/9$.
- 10** Overall multiplier = $0.65 \times 0.9 = 0.585$ **[1]** [3]
 So the customer pays 58.5% of the original, a discount of **41.5% [1]**
 The sign is wrong because the two discounts have been **added**; the second 10% is taken off an already reduced price, so it is worth less than 10% of the original **[1]**

Paper D — Hard

30 MARKS · NON-CALCULATOR

- 1** Profit = $600 - 480 = 120$ **[1]** [3]
 $120 \div 480$ **[1]**
 = **25% profit [1]**
Percentage profit is always measured against the cost price, never the selling price.
- 2** A 20% loss means he received 80%, so the multiplier is 0.8 **[1]** [3]
 $640 \div 0.8$ **[1]**
 = **800 rupees [1]**
- 3** (a) After books he has $2/3$ left. He then spends $1/4$ of that, so he keeps $3/4$ of $2/3$ **[1]** [4]
 $3/4 \times 2/3 = 1/2$ **[1]**
 (b) $1/2$ of the original = 900 **[1]**, so he started with **1800 rupees [1]**
The trap is adding $1/3$ and $1/4$ to get $7/12$ spent. The second fraction is of the remainder, not of the original.

- 4** Fraction removed = $\frac{3}{5} - \frac{1}{3}$ **[1]** [4]
 $= \frac{9}{15} - \frac{5}{15} = \frac{4}{15}$ **[1]**
 $\frac{4}{15}$ of the capacity = 240, so $\frac{1}{15} = 60$ **[1]**
Capacity = $60 \times 15 = \mathbf{900 \text{ litres}}$ **[1]**
- 5** Deal 1: $600 \times 0.8 \times 0.9 = \mathbf{432}$ **[2]** [4]
Deal 2: $600 \times 0.75 \times 0.95 = \mathbf{427.50}$ **[1]**
Deal 2 is better, by 4.50 rupees [1]
Both deals total 30% if you add them, which is exactly why they look identical and are not.
- 6** Take the original as 100; after a 25% rise it is 125 **[1]** [4]
The fall needed is 25 out of 125 **[1]**
 $25 \div 125 = 0.2$ **[1]**
So the price must fall by **20%** **[1]**
Answering 25% scores at most one mark. The fall is measured against the new, larger price — this is the same idea as the 10% up-and-down question on Paper C.
- 7** Girls = $\frac{3}{5}$ of 500 = 300, so boys = 200 **[1]** [4]
25% of 300 = 75 girls walk **[1]**
40% of 200 = 80 boys walk **[1]**
Total = $75 + 80 = \mathbf{155 \text{ students}}$ **[1]**
Averaging 25% and 40% to get 32.5% of 500 gives 162.5 and scores nothing. The two groups are different sizes.
- 8** (i) The mistake: comparing decimals by the size of the digits after the point instead of by place value **[1]** [4]
Padding with a zero gives 0.90 against 0.15, so **0.9 is larger [1]**
(ii) The mistake: adding the numerators and the denominators separately **[1]**
Using a common denominator: $\frac{3}{6} + \frac{2}{6} = \mathbf{\frac{5}{6}}$ **[1]**
A useful check for (ii): the answer must be bigger than $\frac{1}{2}$, and $\frac{2}{5}$ is smaller — so it is wrong before you calculate anything.